EE3110 - Probability Foundations for Electrical Engineers
Tutorial - Week 1

1. If n letters are placed at random in n envelopes, what is the probability that exactly $n-1$ letters will be placed in the correct envelopes?

2. An elevator in a building starts with five passengers and stops at seven floors. If every passenger is equally likely to get off at each floor and all the passengers leave independently of each other, what is the probability that no two passengers will get off at the same floor?

3. Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.

4. You and your friend are playing a game with a standard deck of 52 playing cards. Each card type (2 through 10, Jack, Queen, King, Ace) appears exactly four times in the deck. For this problem, assume the following values for the cards:

   Number cards retain their face value (2 through 10)

   Jack = 11, Queen = 12, King = 13, Ace = 14

   You and your friend each draw one card from the deck without replacement. You draw first, followed by your friend. Whoever draws the card with the higher value wins. In case of a tie (both draw the same value), your friend wins.

   What is the probability that you win this game?

5. Let A and B be two events such that $\boldsymbol{P}(A) > \boldsymbol{P}(B) > 0$ and $\boldsymbol{P}(A) + \boldsymbol{P}(B) \geq 1$.
   (a) Determine upper and lower bounds on $\boldsymbol{P}(A \cap B)$ in terms of $\boldsymbol{P}(A), \boldsymbol{P}(B)$.
   (b) When $\boldsymbol{P}(A) \geq 1-\delta$ and $\boldsymbol{P}(B) \geq 1-\delta$, express the lower bound on $\boldsymbol{P}(A \cap B)$ in terms of δ.

6. There are 2^k tennis players. The players are of equal ability, i.e. in a match between any two players, probability of each of them winning is $\frac{1}{2}$.

   The players play an elimination tournament with random initial seeding. That is, the players are arranged randomly, then 2^{k-1} matches are constituted among these players such that seed 1 plays seed 2 in match 1, seed 3 plays seed 4 in match 2, and so on. The winners of these matches go to the next round which would have 2^{k-1} players and 2^{k-2} rounds (winner of match 1 plays winner of match 2), and so on until there is a single winner.

   1. What is the probability that any two given players meet in the first round? $\frac{1}{2^k-1}$
   2. What is the probability that any two given players have a match during the final? $\frac{1}{2^k C_2}$
   3. What is the probability that any two given players have a match during the tournament? $\frac{1}{2^k-1}$

7. Sixteen players, $P_1, P_2, \ldots, P_{16}$, compete in a knockout tournament. It is known that whenever players P_i and P_j face each other, P_i will win if $i < j$. Assuming the players are paired randomly in each round, what is the probability that player P_4 reaches the semi-finals?

8. You roll a fair 6-sided die n times. Find the probability that all 6 sides appear at least once. Also calculate the probability that exactly 4 of the 6 sides appear among the n rolls.

9. Each of k jars contains m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally, a ball is randomly chosen from jar k. Show that the probability that the last ball is white is the same as the probability that the first ball is white, i.e., it is $m/(m+n)$.

Jar 2: $\frac{m}{(m+n)} \times \frac{m+1}{(m+n+1)} + \frac{n}{(m+n)} \times \frac{m}{(m+n+1)}$

$\frac{m(m+n+1)}{(m+n)(m+n+1)} = \frac{m}{m+n}$

$$\frac{1}{r+w C_r} = \frac{r!\,w!}{(r+w)!}$$

10. Suppose that a box contains r red balls, w white balls, and b blue balls. Suppose also that balls are drawn from the box one at a time, at random, without replacement. What is the probability that all r red balls will be obtained before any white balls are obtained?

11. What is the smallest number k such that in a group of k people, there is a greater than 50% chance that at least two people share the same birthday? (Assume there are exactly 365 days in a year and each day is equally likely to be someone's birthday. Use the approximation $\ln(1-x) \approx -x$ for small x and $\ln(e^x) = x$)

12. I bought two fair six-sided dice. Keeping the first one untouched (numbered 1 through 6), I modified the second die by erasing all its faces and labeling them with real numbers of my choice (not necessarily distinct). After rolling both dice together, I observed that the sum of the two outcomes is equally likely to be any integer from 1 to 12. Determine the numbers written on the modified die.

13. **Circular Equilibrium and Stability:**
A circular platform floats frictionlessly in space. You place N identical point masses on the circumference of the platform. Each mass is placed independently and uniformly at random. The system is said to be in a *stable configuration* if there exists a rotation of the platform such that all the masses lie within a semicircle (i.e., within an arc of length π). What is the probability that the configuration is stable?

14. **(Something to think about)** Assume that there is a roulette wheel, such that it is equally likely to land at any particular angle. The probability of getting a particular angle (say 30°) is 0. However, once the roulette wheel is spun, it would have to land somewhere (say it lands at 30°). Thus, a probability 0 event has occurred. Is this a discrepancy?

It depends on the least count of the wheel markings.
The probability is not 0 but close to 0

1. If n letters are placed at random in n envelopes, what is the probability that exactly $n-1$ letters will be placed in the correct envelopes?

0 (∵ it's impossible to have exactly n-1 right)

2. An elevator in a building starts with five passengers and stops at seven floors. If every passenger is equally likely to get off at each floor and all the passengers leave independently of each other, what is the probability that no two passengers will get off at the same floor?

$$\frac{{}^{7}C_{5} \times 5!}{7^5}$$

3. Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.

A → genius B → chocolate lover

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 + P(A \cap B) - P(A) - P(B)$$

$$= 1 + \frac{4}{10} - \frac{6}{10} - \frac{7}{10} = \frac{1}{10}$$

4. You and your friend are playing a game with a standard deck of 52 playing cards. Each card type (2 through 10, Jack, Queen, King, Ace) appears exactly four times in the deck. For this problem, assume the following values for the cards:

   Number cards retain their face value (2 through 10)

   Jack = 11, Queen = 12, King = 13, Ace = 14

   You and your friend each draw one card from the deck without replacement. You draw first, followed by your friend. Whoever draws the card with the higher value wins. In case of a tie (both draw the same value), your friend wins.

   What is the probability that you win this game?

$$4/52 \left[0 + 4/51 + 8/51 + \cdots + 48/51\right] = 4/\cancel{52}^{13} \cdot \frac{\cancel{6}^{2} \cdot 12}{51_{17}} = 8/17$$

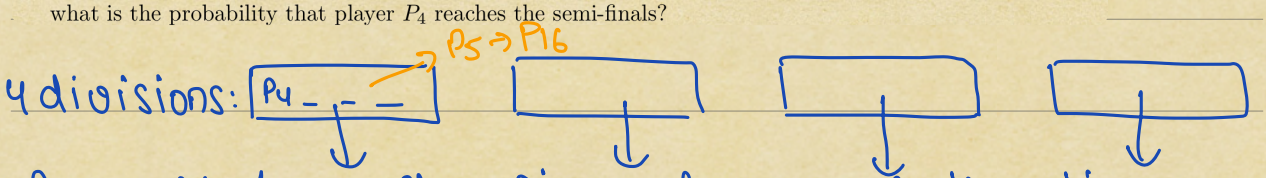
5. Let A and B be two events such that $\boldsymbol{P}(A) > \boldsymbol{P}(B) > 0$ and $\boldsymbol{P}(A) + \boldsymbol{P}(B) \geq 1$.
   (a) Determine upper and lower bounds on $\boldsymbol{P}(A \cap B)$ in terms of $\boldsymbol{P}(A), \boldsymbol{P}(B)$.
   (b) When $\boldsymbol{P}(A) \geq 1 - \delta$ and $\boldsymbol{P}(B) \geq 1 - \delta$, express the lower bound on $\boldsymbol{P}(A \cap B)$ in terms of δ.

(a) $P(A \cap B) = P(A) + P(B) - P(A \cup B) = P(B)$ upper bound

$P(A) + P(B) - 1$ lower bound

(b) $P(A \cap B) \geq P(A) + P(B) - 1 \geq 1 - 2\delta$

7. Sixteen players, $P_1, P_2, \ldots, P_{16}$, compete in a knockout tournament. It is known that whenever players P_i and P_j face each other, P_i will win if $i < j$. Assuming the players are paired randomly in each round, what is the probability that player P_4 reaches the semi-finals?



P_4 must be a champion of one of the div

$$\Rightarrow \frac{{}^{12}C_3}{{}^{15}C_3} = 44/91$$

$\to P_5 \to P_{16}$

$\to P_1 \to P_3$, $P_5 \to P_{16}$

8. You roll a fair 6-sided die n times. Find the probability that all 6 sides appear at least once. Also calculate the probability that exactly 4 of the 6 sides appear among the n rolls.

inclusion-exclusion

1-P(atleast one missed)

a) $$\frac{6^n - {}^6C_1 5^n + {}^6C_2 4^n - {}^6C_3 3^n + {}^6C_4 2^n - {}^6C_5 1^n}{6^n} = P(\text{all } 6)$$

(1 miss, 2 miss, 3 miss, 4 miss, 5 miss)

b) $$\frac{\left(4^n - {}^4C_1 3^n + {}^4C_2 \cdot 2^n - {}^4C_3 \cdot 1^n\right) {}^6C_4}{6^n} = P(\text{exactly } 4)$$

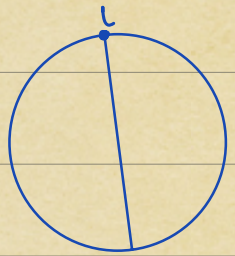
(1 miss, 2 miss, 3 miss; select 4)

13. **Circular Equilibrium and Stability:**
A circular platform floats frictionlessly in space. You place N identical point masses on the circumference of the platform. Each mass is placed independently and uniformly at random. The system is said to be in a *stable configuration* if there exists a rotation of the platform such that all the masses lie within a semicircle (i.e., within an arc of length π). What is the probability that the configuration is stable?

$E_i \to i^{th}$ point is leading

$$P = P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i) = n/2^{n-1}$$

$P(E_i) = 1/2 \cdot 1/2 \cdots = 1/2^{n-1}$



6. There are 2^k tennis players. The players are of equal ability, i.e. in a match between any two player probability of each of them winning is $\frac{1}{2}$.

   The players play an elimination tournament with random initial seeding. That is, the players ar arranged randomly, then 2^{k-1} matches are constituted among these players such that seed 1 plays see 2 in match 1, seed 3 plays seed 4 in match 2, and so on. The winners of these matches go to the nex round which would have 2^{k-1} players and 2^{k-2} rounds (winner of match 1 plays winner of match 2 and so on until there is a single winner.

   1. What is the probability that any two given players meet in the first round?
   2. What is the probability that any two given players have a match during the final?
   3. What is the probability that any two given players have a match during the tournament?

2^{k-1} matches in round 1

(a) $\frac{2^{k-1}}{(2^k C_2)}$ → they should be paired in one of 2^{k-1} matches
→ total pairs of players

(b) $\frac{1}{(2^k C_2)}$ → favorable
→ total pairs

(c) total games $= 2^k - 1$

$P(\text{round } 1) = \frac{2^{k-1}}{2^k C_2}$

$P(\text{round } 2) = \frac{2^{k-2}}{2^k C_2} \Rightarrow P = \left(\frac{2^k - 1}{2^k C_2}\right)$

$\vdots$

$P(\text{round } k) = \frac{1}{2^k C_2}$

11. What is the smallest number k such that in a group of k people, there is a greater than 50% chance that at least two people share the same birthday? (Assume there are exactly 365 days in a year and each day is equally likely to be someone's birthday. Use the approximation $\ln(1-x) \approx -x$ for small x and $\ln(e^x) = x$)

P(no shared birth day) =

$\frac{365}{365} \times \frac{364}{365} \times \cdots \frac{366-k}{365} < \frac{1}{2}$

$$\Rightarrow \frac{\prod_{i=1}^{k}(366-i)}{365^k} < \frac{1}{2}$$

$$\Rightarrow \sum_{i=0}^{k-1} \ln\left(\frac{365-i}{365}\right) < \ln(1/2)$$

$$\Rightarrow -\sum_{i=0}^{k-1} \frac{i}{365} < \ln(1/2)$$

$$\Rightarrow \frac{(k-1)k}{730} > \ln 2 \Rightarrow k^2 \approx 730 \ln 2 \Rightarrow k \approx 22.5$$

$$\Rightarrow k_{min} = 23$$

12. I bought two fair six-sided dice. Keeping the first one untouched (numbered 1 through 6), I modified the second die by erasing all its faces and labeling them with real numbers of my choice (not necessarily distinct). After rolling both dice together, I observed that the sum of the two outcomes is equally likely to be any integer from 1 to 12. Determine the numbers written on the modified die.

D_1	1 (1/6)	2 (1/6)	3 (1/6)	4 (1/6)	5 (1/6)	6 (1/6)
D_2	a (1/6)	b (1/6)	c (1/6)	d (1/6)	e (1/6)	f (1/6)

$(a \le b \le c \le d \le e \le f)$

3 pairs for sum = k

$1+a=1$; $6+f=12 \Rightarrow a=0, f=6$

$2+a=1$ X or $1+b=1 \Rightarrow b=0$

$(0,0,0,6,6,6)$

EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 2

1. Suppose a test for COVID is 99% accurate in both directions (positive and negative) and 0.3% of the population is COVID positive. If someone tests positive, what is the probability they actually are COVID positive? Justify why you are getting a low probability even though the test appears to be very accurate.

2. We throw 3 dice one by one. What is the probability that we obtain 3 numbers in strictly increasing order?

3. A coin is tossed twice. Harshith claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is he right? Does it make a difference if the coin is fair or unfair? How can we generalize Harshith's reasoning?

4. Let A and B be events with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. We say that an event B *suggests* an event A if $\mathbb{P}(A|B) > \mathbb{P}(A)$, and does not suggest event A if $\mathbb{P}(A|B) < \mathbb{P}(A)$.

   1. Show that B suggests A if and only if A suggests B.
   2. Show that B suggests A if and only if B^c does not suggest A. Assume that $\mathbb{P}(B^c) > 0$.
   3. We know that a treasure is located in one of two places, with probabilities β and $1-\beta$, respectively, where $0 < \beta < 1$. We search the first place and if the treasure is there, we find it with probability $p > 0$. Show that the event of not finding the treasure in the first place suggests that the treasure is in the second place.

5. For events A, B and C with $C \subseteq B$ and $P(A)P(B)P(C) > 0$, $P(A|B) = \alpha P(A|B \cap C) + (1-\alpha)P(A|B \cap C^c)$. Find α in terms of $P(A), P(B)$ and $P(C)$.

6. A large class in probability theory is taking a multiple-choice test. For a particular question on the test, the fraction of examinees who know the answer is p; $1-p$ is the fraction that will guess. The probability of answering a question correctly is unity for an examinee who knows the answer and $1/m$ for a guesser; m is the number of multiple choice alternatives. Compute the probability that an examinee knew the answer to a question, given that he or she has answered it correctly.

7. *The prisoner's dilemma*: The release of two out of three prisoners has been announced, but their identity is kept secret. One of the prisoners considers asking a friendly guard to tell him who is the prisoner other than himself that will be released, but hesitates based on the following rationale: at the prisoner's present state of knowledge, the probability of being released is $\frac{2}{3}$, but after he knows the answer, the probability of being released will become $\frac{1}{2}$, since there will be two prisoners (including himself) whose fate is unknown and exactly one of the two will be released. What is wrong with this line of reasoning?

8. **(a)** Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.

   Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?

   (b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n-2$.

1. Suppose a test for COVID is 99% accurate in both directions (positive and negative) and 0.3% of the population is COVID positive. If someone tests positive, what is the probability they actually are COVID positive? Justify why you are getting a low probability even though the test appears to be very accurate.

$$P(+ve/got\ +ve) = \frac{3/1000 \times 99/100}{3/1000 + 99/100 + 997/1000 \times 1/100}$$

$$= \frac{3\times 99}{3\times 99 + 997} = \frac{297}{1294}$$

3. A coin is tossed twice. Harshith claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is he right? Does it make a difference if the coin is fair or unfair? How can we generalize Harshith's reasoning?

$P(\text{head}) = p \Rightarrow P(\text{two heads}) = p^2$

$P(2\ \text{Head} / 1^{st}\ \text{is head}) = \frac{1 \times p^2}{p} = p$

$P(2\ \text{head} / \text{atleast 1 is head}) = \frac{1 \times p^2}{2p(1-p)+p^2} = \frac{p^2}{2p-p^2} = \frac{p}{2-p}$

$p \geq \frac{p}{2-p}$ (true)

4. Let A and B be events with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. We say that an event B *suggests* an event A if $\mathbb{P}(A|B) > \mathbb{P}(A)$, and does not suggest event A if $\mathbb{P}(A|B) < \mathbb{P}(A)$.

   1. Show that B suggests A if and only if A suggests B.
   2. Show that B suggests A if and only if B^c does not suggest A. Assume that $\mathbb{P}(B^c) > 0$.
   3. We know that a treasure is located in one of two places, with probabilities β and $1-\beta$, respectively, where $0 < \beta < 1$. We search the first place and if the treasure is there, we find it with probability $p > 0$. Show that the event of not finding the treasure in the first place (B) suggests that the treasure is in the second place. (A)

(a) $P(A/B) > P(A) \Rightarrow P(A \cap B) > P(A)\,P(B)$

$\Rightarrow P(A \cap B)/P(A) > P(B) \Rightarrow P(B/A) > P(B)$

(b) $P(A/B) > P(A) \Rightarrow P(A \cap B) > P(A)\,P(B)$ $\quad \frac{1-\beta}{\beta(1-p)+(1-\beta)}$

$\Rightarrow P(A) - P(A \cap B^c) > P(A) - P(A)P(B^c)$

$\Rightarrow P(A \cap B^c) < P(A)P(B^c)$

$\Rightarrow P(A|B^c) < P(A)$

(c) A: not finding treasure in 1st place $P(A) = 1-p$

B: treasure in second place $P(B) = 1-\beta$

$P(B|A) > P(B)$ $\quad \frac{1-\beta}{-\beta p+1} > 1-\beta \Rightarrow 1 > 1-\beta p$ ✓

5. For events A, B and C with $C \subseteq B$ and $P(A)P(B)P(C) > 0$, $P(A|B) = \alpha P(A|B \cap C) + (1-\alpha)P(A|B \cap C^c)$. Find α in terms of $P(A), P(B)$ and $P(C)$.

$\frac{P(A \cap B)}{P(B)} = \alpha \frac{P(A \cap B \cap C)}{P(B \cap C)} + (1-\alpha) \frac{P(A \cap B \cap C^c)}{P(B \cap C^c)}$

$P(B \cap C) \to P(C)$

$P(B \cap C^c) \to P(B) - P(C)$

$\frac{P(A \cap B \cap C) + P(A \cap B \cap C^c)}{P(B)} = \alpha \frac{P(A \cap B \cap C)}{P(C)} + (1-\alpha) \frac{P(A \cap B \cap C^c)}{P(B) - P(C)}$

$\Rightarrow \alpha = \frac{P(C)}{P(B)} = P(C|B)$

6. A large class in probability theory is taking a multiple-choice test. For a particular question on the test, the fraction of examinees who know the answer is p; $1-p$ is the fraction that will guess. The probability of answering a question correctly is unity for an examinee who knows the answer and $1/m$ for a guesser; m is the number of multiple choice alternatives. Compute the probability that an examinee knew the answer to a question, given that he or she has answered it correctly.

$P(A|B) \overset{A}{=} \frac{P(B|A)P(A)}{P(B)} \overset{B}{=} \frac{1 \cdot p}{p + (1-p)/m} = \frac{pm}{mp+1-p}$

7. *The prisoner's dilemma*: The release of two out of three prisoners has been announced, but their identity is kept secret. One of the prisoners considers asking a friendly guard to tell him who is the prisoner other than himself that will be released, but hesitates based on the following rationale: at the prisoner's present state of knowledge, the probability of being released is $\frac{2}{3}$, but after he knows the answer, the probability of being released will become $\frac{1}{2}$, since there will be two prisoners (including himself) whose fate is unknown and exactly one of the two will be released. What is wrong with this line of reasoning?

The prisonner's reasoning is flawed because his new info doesn't change the total number of prisonners being released, but it rather changes the outcomes he considers.

8. **(a)** Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.

   Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?

   (b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n-2$.

P(chosen door has car) = $1/n$

P(car is behind one of other doors) = $\frac{n-1}{n}$

$n-1-m$ doors left to switch

P(success after switching) = $\frac{\frac{n-1}{n}}{n-m-1} = \frac{n-1}{n(n-1-m)}$

P(success without switching) = $\frac{1}{n}$ (no change)

$\frac{n-1}{n(n-m-1)} > \frac{1}{n} \Rightarrow$ always switch

9. To encourage Elmer's promising tennis career, his father offers him a prize if he wins (at least) two tennis sets in a row in a three-set series to be played with his father and the club champion alternately: father–champion–father or champion–father–champion, according to Elmer's choice. The champion is a better player than Elmer's father. Which series should Elmer choose?

10. Alice searches for her term paper in her filing cabinet, which has several drawers. She knows that she left her term paper in drawer j with probability $p_j > 0$. The drawers are so messy that even if the paper is present in drawer i, the probability that she finds it is only d_i.

    Alice unsuccessfully searches in a particular drawer i. Conditioned on this event, find the probability that the paper is in drawer j for: (a) $j \neq i$, (b) $j = i$.

11. You are taking out candies one by one from a jar that has 10 red candies, 20 blue candies, and 30 green candies in it. What is the probability that there are at least 1 blue candy and 1 green candy left in the jar when you have taken out all the red candies?

12. Consider a coin that comes up heads with probability p and tails with probability $1 - p$. Let q_n be the probability that after n independent tosses, there have been an even number of heads. Derive a recursion that relates q_n to q_{n-1}, and solve this recursion to establish the formula

$$q_n = \frac{1 + (1 - 2p)^n}{2}.$$

13. A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n-th passenger in line has a ticket for seat number n. Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their assigned seats unless it is already occupied; If it is occupied, they will then find a free seat to sit in, at random. What is the probability that the last (100th) person to board the plane will sit in their own seat?

14. There are N urns of which the r-th urn contains $r - 1$ red balls and $N - r$ blue balls. You pick an urn at random and remove two balls at random without replacement. Find the probability that

    (a) the second ball is blue.

    (b) the second ball is blue, given that the first is blue.

15. **Gambler's Ruin.** Suppose you play a game by tossing a fair coin repeatedly and independently. If it comes up heads, you win a dollar, and if it comes up tails, you lose a dollar. Suppose you start with \$50. What is the probability you will get to \$200 without first getting ruined (running out of money)?

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C-F-C $\qquad P_F > P_C$

$$P(C-F) + P(\bar{C}-F-C) = P_C P_F + (1-P_C) P_F P_C$$

$$= P_C P_F (2-P_C)$$

F-C-F

$$P(F-C) + P(\bar{F}-C-F) = P_F P_C + (1-P_F) P_C P_F$$

$$= P_C P_F (2-P_F)$$

C-F-C $\Rightarrow$ better

11. You are taking out candies one by one from a jar that has 10 red candies, 20 blue candies, and 30 green candies in it. What is the probability that there are at least 1 blue candy and 1 green candy left in the jar when you have taken out all the red candies?

$$1 - P(\text{no B}) - P(\text{no G}) + P(\text{no B \& G})$$

$$1 - \frac{1}{3}\left(\frac{10}{10+20}\right) - \frac{1}{4}\left(\frac{10}{10+30}\right) + \frac{1}{6}\left(\frac{10}{10+20+30}\right) = \frac{7}{12}$$

14. There are N urns of which the r-th urn contains $r-1$ red balls and $N-r$ blue balls. You pick an urn at random and remove two balls at random without replacement. Find the probability that

(a) the second ball is blue.

(b) the second ball is blue, given that the first is blue.

(a) $P(2^{nd} \text{ is blue}) = \sum P(2^{nd} \text{ is blue} \mid \text{urn } r) \cdot P(\text{urn } r)$

$$= \sum_{r=1}^{N} \frac{N-r}{N-1} \times \frac{1}{N} = \frac{(N-1)N}{2N(N-1)} = \frac{1}{2}$$

(b) $P(2^{nd} \text{ is blue} \mid 1^{st} \text{ is blue}) = \dfrac{P(2^{nd} \cap 1^{st})}{P(1^{st}) \to 1/2}$

$$\frac{1/N \sum_{r=1}^{N} \frac{(N-r)}{(N-1)} \frac{(N-r-1)}{(N-2)}}{1/2} = \frac{2}{N(N-1)(N-2)} \sum_{t=0}^{N-1} t(t-1)$$

$$= \frac{2}{N(N-1)(N-2)} \left[\frac{(N-1)(N)(2N-1)}{6} - \frac{(N-1)N}{2} \right]$$

$$= 1/3(N-2) \left[(2N-1) - 3 \right] = 2/3$$

12. Consider a coin that comes up heads with probability p and tails with probability $1-p$. Let q_n be the probability that after n independent tosses, there have been an even number of heads. Derive a recursion that relates q_n to q_{n-1}, and solve this recursion to establish the formula

$$q_n = \frac{1 + (1-2p)^n}{2}.$$

$q_n = (q_{n-1})(1-p) + (1-q_{n-1})p = (q_{n-1})(1-2p) + p$

$q_0 = 1$

homogeneous part: $q_n^h = q_{n-1}^h (1-2p)$

$\Rightarrow q_n^h = C(1-2p)^n$

particular solution: $A = (1-2p)A + p \Rightarrow A = 1/2$

$q_n = q_n^h + q_n^p = q_n^h + A \quad q_0 = 1 \Rightarrow C = \frac{1}{2} \Rightarrow$

$q_n = \frac{1}{2} + \frac{1}{2}(1-2p)^n$

13. A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n-th passenger in line has a ticket for seat number n. Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their assigned seats unless it is already occupied; If it is occupied, they will then find a free seat to sit in, at random. What is the probability that the last (100th) person to board the plane will sit in their own seat?

if drunk guy takes his seat $\Rightarrow$ 100th takes 100-seat

" " " " 100 seat ⇒ 100th doesn't get his seat at all

each other passenger is equally likely to take seat #n or seat #100. $p = 1/2$

15. **Gambler's Ruin.** Suppose you play a game by tossing a fair coin repeatedly and independently. If it comes up heads, you win a dollar, and if it comes up tails, you lose a dollar. Suppose you start with \$50. What is the probability you will get to \$200 without first getting ruined (running out of money)?

$P(K) = p\,P(K+1) + (1-p)\,P(K-1)$ $\quad P(0) = 0$, $P(n) = 1$

$r^K = p \cdot r^{K+1} + (1-p)\, r^{K-1}$

$\Rightarrow p r^2 - r + (1-p) = 0 \Rightarrow r = \frac{1 \pm (1-2p)}{2p} = 1, \frac{1-p}{p}$

$P(K) = a_1 \cdot + a_2 \left(\frac{1-p}{p}\right)^K = \frac{(q/p)^K - 1}{(q/p)^n - 1}$ $\quad (q = 1-p)$

$P(K) = \frac{K}{N}$ (if $q = p$)

EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 3

1. Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, etc. (so X takes values 1, 2..., 7, with equal probabilities). Let Y be the next day after X (again represented as an integer between 1 and 7). Do X and Y have the same distribution? What is $P(X < Y)$?

2. Let X and Y be two independent $Ber(\frac{1}{2})$ random variables. Define random variables U and V by:

$$U = X + Y \text{ and } V = |X - Y|$$

   1. Determine the joint and marginal probability distributions of U and V.
   2. Find out whether U and V are dependent or independent.

3. Let X be a random variable with law $f(x) = C\frac{2^x}{x!}$ for $x \in \mathbb{N}$. Then, the probability that X is even is approximately _____.

4. **Happy Birthday again!** You go to a party with 500 guests. What is the probability that exactly one other guest has the same birthday as you? Calculate this exactly and also approximately by using the Poisson PMF. (For simplicity, exclude birthdays on February 29.)

5. **Memorylessness** Let X be a discrete random variable following the Geometric PMF with parameter p. Show the following:

$$\mathbb{P}(X = n + k | X > n) = \mathbb{P}(X = k). \tag{1}$$

   This means that given you have a already waited for n trails for a success the probability that you will wait for k more trails is same as the probability of waiting for k trails from the first trail, i.e., the random variable doesn't remember the past.

6. The joint probability distribution of two discrete random variables X and Y is given by:

$X \backslash Y$	y_1	y_2	y_3
x_1	0.10	0.20	0.10
x_2	0.15	0.25	0.20

   1. **Marginal distributions**
      (a) Find the marginal distribution of X.
      (b) Find the marginal distribution of Y.
   2. **Conditional distributions**
      (a) Compute $P(X = x_1 \mid Y = y_2)$.
      (b) Compute $P(Y = y_3 \mid X = x_2)$.
   3. **Theorem of Total Probability (verification)**
      Verify that
$$P(X = x_i) = \sum_j P(X = x_i \mid Y = y_j)\, P(Y = y_j)$$
      by carrying out the calculation explicitly for $X = x_1$.

7. Alice flips a fair coin n times and Bob flips another fair coin $n + 1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n + 1, \frac{1}{2})$.
   (a) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.

3. Let X be a random variable with law $f(x) = C\frac{2^x}{x!}$ for $x \in \mathbb{N}$. Then, the probability that X is even is approximately _____.

$f(x) = C\frac{2^x}{x!} \quad x \in N \quad P(X = even)$

$\sum_{x=1}^{\infty} \frac{2^x}{x!} = \frac{1}{C} \Rightarrow C = \frac{1}{e^2-1}$

$P(X = 2K;\ K \in N) = \frac{1}{e^2-1} \sum_{x=1}^{\infty} \frac{2^{2x}}{(2x)!} = \frac{1}{e^2-1}\left(\frac{e^2+e^{-2}}{2} - 1\right)$

$= \frac{1}{(e^2-1)} \frac{(e-1/e)^2}{2} = \frac{(e^2-1)}{2e^2}$

5. **Memorylessness** Let X be a discrete random variable following the Geometric PMF with parameter p. Show the following:

$$\mathbb{P}(X = n + k | X > n) = \mathbb{P}(X = k). \tag{1}$$

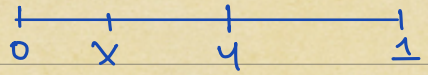
This means that given you have a already waited for n trails for a success the probability that you will wait for k more trails is same as the probability of waiting for k trails from the first trail, i.e., the random variable doesn't remember the past.

$P(X = K) = p(1-p)^{K-1} \qquad P(X = n+K \mid X > n) = P(X = K)$

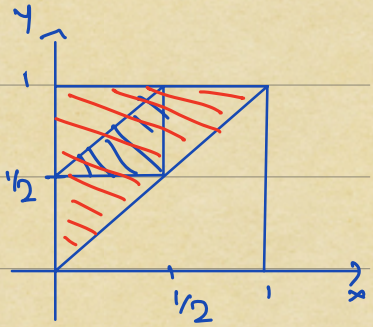
$\frac{P(X = n+K)}{P(X > n)} = \frac{p(1-p)^{n+K-1}}{p(1-p)^n + p(1-p)^{n+1} + \dots}$

$= \frac{(1-p)^{n+K-1}}{(1-p)^n \frac{1}{1-(1-p)}} = p(1-p)^{K-1}$

8. **Probability of Triangle** Two points are chosen with uniform probability and independently on a stick. It is broken at those two points to form three segments. What is the probability that the three segments can form a triangle?



$x,\ y-x,\ 1-y$



$x+(y-x) > 1-y \Rightarrow y > 1/2$

$(y-x)+(1-y) > x \Rightarrow x < 1/2$

$x+(1-y) > y-x \Rightarrow y < x+1/2$

$P = \text{area(blue)} / \text{area(red)} = 1/4$

(b) Compute $P(X < Y)$.

8. **Probability of Triangle** Two points are chosen with uniform probability and independently on a stick. It is broken at those two points to form three segments. What is the probability that the three segments can form a triangle?

9. Let $X_1, X_2, \ldots, X_n$ be i.i.d. real-valued random variables with common CDF $F(x) = \mathbb{P}(X_1 \leq x)$. Define the maximum $M = \max_{1 \leq i \leq n} X_i$ and the minimum $m = \min_{1 \leq i \leq n} X_i$. Derive the CDFs $F_M(x) = \mathbb{P}(M \leq x)$ and $F_m(x) = \mathbb{P}(m \leq x)$.

10. A packet communication system consists of a buffer that stores packets from some source and a communication line that retrieves packets from the buffer and transmits them to a receiver. The communication system operates in time-slot pairs. In the first slot, the system stores some packets that are generated by the source according to a Poisson PMF with parameter λ. However, the maximum number of packets that can be stored is b (some integer), and packets arriving after the buffer is full are discarded. In the second slot, the system transmits either all the stored packets or c $(0 < c < b)$ packets (whichever is less).

    1. Assuming that at the beginning of the first slot the buffer is empty, find the PMF of the number of packets stored at the end of the first slot and at the end of the second slot.
    2. What is the probability that some packets get discarded during the first slot?

11. *(Buffon's Needle)* If a short needle of length l is dropped on a paper that is ruled with equally spaced lines of distance $d \geq l$, find the probability that the needle comes to lie in a position where it crosses one of the lines.
(Some trivia : This result was used as a measure of the value of pi experimentally in the 18th century.)

12. Arjun, Nishanth and Harshith are roommates. Every day they promise each other that they will sleep at 1:00 AM, but end up having life discussions, so they always sleep late.

    - The time they **go to sleep** S can be modeled as a uniformly distributed **continuous** random variable between 1:00 AM and 6:00 AM, that is, $S \sim \text{Uniform}(1, 6)$.
    - The time they **wake up** W can be modeled as a uniformly distributed **continuous** random variable between 8:00 AM and 12:00 PM, that is, $W \sim \text{Uniform}(8, 12)$.

    Let us assume that S and W are independent.

    (a) Every night, Arjun confidently proclaims that he will wake up at **exactly** 8:30 am and have breakfast. What is the probability that this happens?

    (b) They have their first class at 9:00 AM. They will go to class if they wake up at or before 9:00 AM. What is the probability that they will go to class?

    (c) They can stay awake in class if they get a minimum of 6 hours of sleep (basically just enough sleep). What is the probability that they will stay awake?

    (d) DDiet is an eatery outside T Gate. If they are awake at 4:00 A.M. then they will go to DDiet to have early morning tea with a probability of 0.7. What is the probability that they go to DDiet and also have enough sleep?